

Clustering and Differential Analysis

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Roadmap: From biology to methodology

14: Biological Networks

- Protein-protein, genetic interactions
- Cell-cell interaction networks
- Network as a tool: propagation

15: Clustering

- Graph-based clustering
- When clustering goes wrong

Today's lecture

- 1 Graph-based clustering
- 2 Single-cell clustering in practice
- 3 When clustering goes wrong

Last time: networks encode relationships

- We learned how to build and visualize networks
- Protein-protein interactions, genetic interactions, cell-cell communication
- Today: how do we find **groups** within a network?

Recap: igraph in R

```
library(igraph)
```

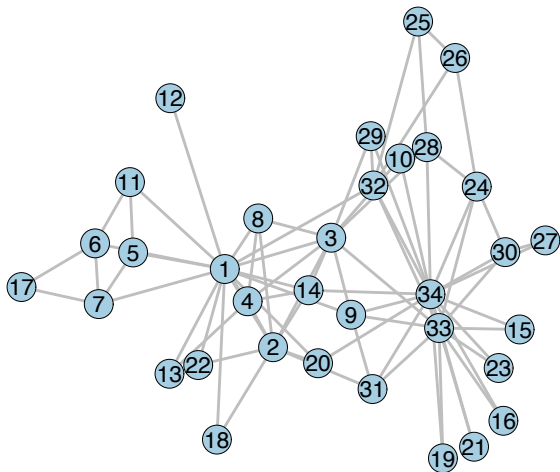
```
karate <- .read.online("http://www-personal.umich.edu/~mejn/netdata/karate.zip")
karate
```

```
## IGRAPH b851c8c U--- 34 78 --
## + attr: id (v/n)
## + edges from b851c8c:
## [1] 1-- 2 1-- 3 2-- 3 1-- 4 2-- 4 3-- 4 1-- 5 1-- 6 1-- 7 5-- 7
## [11] 6-- 7 1-- 8 2-- 8 3-- 8 4-- 8 1-- 9 3-- 9 3--10 1--11 5--11
## [21] 6--11 1--12 1--13 4--13 1--14 2--14 3--14 4--14 6--17 7--17
## [31] 1--18 2--18 1--20 2--20 1--22 2--22 24--26 25--26 3--28 24--28
## [41] 25--28 3--29 24--30 27--30 2--31 9--31 1--32 25--32 26--32 29--32
## [51] 3--33 9--33 15--33 16--33 19--33 21--33 23--33 24--33 30--33 31--33
## [61] 32--33 9--34 10--34 14--34 15--34 16--34 19--34 20--34 21--34 23--34
## [71] 24--34 27--34 28--34 29--34 30--34 31--34 32--34 33--34
```

- $n = |V|$: number of vertices, $m = |E|$: number of edges
- d_v : degree of vertex v (how many neighbours?)

A classic example: Karate club (Zachary 1977)

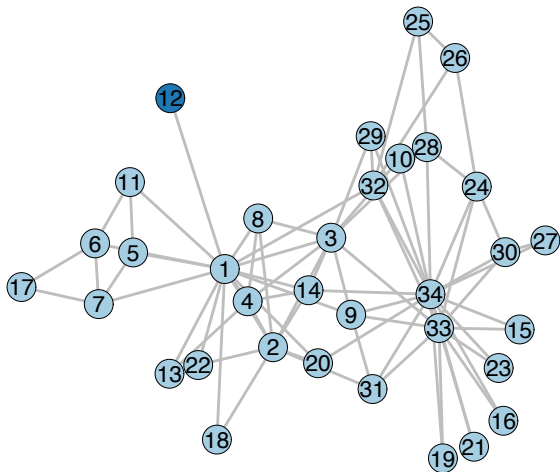
- 34 members, edges = friendships
- The club split into two factions after a dispute
- Can we recover this split from the network alone?
- Who are the “influencers”?
 - **1**: the instructor (“Mr. Hi”)
 - **34**: the president (John A.)



Idea: split the network by cutting edges

- Consider two groups — find a set of edges to cut so the graph splits into two disconnected components
- **Min-cut**: the fewest edges to remove

A trivial solution?



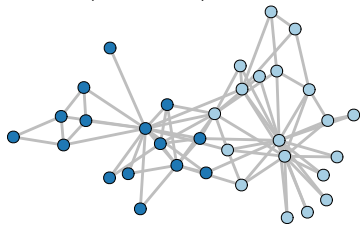
Split G by the fewest edges between two parties

```
.cuts <-  
  as.directed(karate) %>%  
  st_min_cuts(source = "1", target = c("33", "34"))  
.cuts$value / 2  # number of edges to cut
```

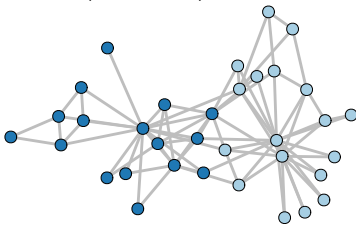
```
## [1] 5
```


Min-cut partitions from source to target

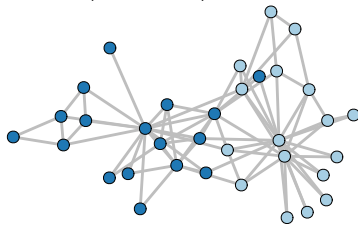
Partition 1 (15 members)



Partition 2 (16 members)



Partition 3 (17 members)



- Multiple solutions with the same min-cut = 5 edges
- But min-cut requires knowing **source and target** — can we find groups without labels?

The network as an adjacency matrix



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- Row/column: a member
- Non-zero element: friendship
- Can you see group structures?

A generative approach: Stochastic block model

- View adjacency matrix A as **data**, not just structure
- **Model:** $A_{ij} \sim \text{Bernoulli}(\theta_{k(i),k(j)})$
 - Each vertex belongs to a latent cluster
 - Edge probability depends only on cluster membership
- **Learn:**
 - $Z_{ki} = 1$ if vertex i in cluster k (cluster $\times$ vertices)
 - $\Theta_{kk'}$ = edge probability between clusters k and k'
- **Goal:** factorize $A \approx Z^\top \Theta Z$

Fitting the stochastic block model

```
library(hsblock)
.out <- fit.hsblock(A, "bernoulli", vbiter=100,
                    inner.iter=10, verbose=FALSE)
Z <- as.matrix(.out$Z)
```

```
n <- ncol(Z)
Edges <- 0.5 * Z %*% A %*% t(Z)
.ones <- matrix(1,n,n)
diag(.ones) <- 0
Tot <- Z %*% .ones %*% t(Z)/2
Theta <- Edges / Tot
```

- Z : 2 clusters $\times$ 34 vertices
- Θ : edge probabilities

Matrix factorization view: $A \approx Z^T \Theta Z$

A (34×34)



$\approx$

Z'



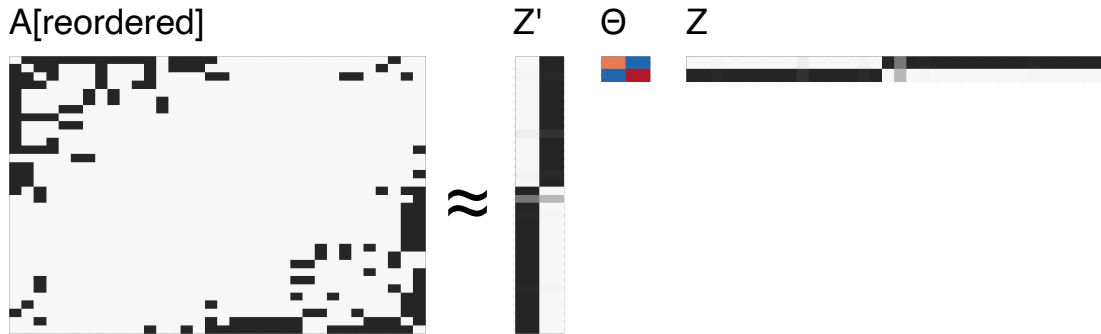
Θ



Z



Reordering by cluster reveals the block structure



- Sorting vertices by cluster membership makes the block structure visible
- Each block shows edge density within/between groups

What is modularity? (Newman and Girvan 2004)

Given a partition of vertices into groups, **modularity** Q measures how many more edges fall *within* groups than expected by chance:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{d_i d_j}{2m} \right] \delta(c_i, c_j)$$

- A_{ij} : adjacency matrix, d_i : degree of vertex i , m : total edges
- $\delta(c_i, c_j) = 1$ if vertices i, j are in the same group
- $Q \in [-0.5, 1]$; higher = stronger community structure
- The null model: edges are distributed proportionally to degree

Greedy optimization

(Blondel et al. 2008; Traag et al. 2019)

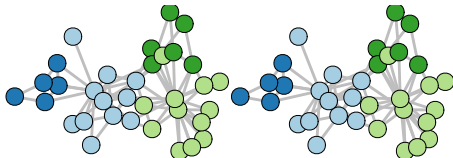
Louvain (2008): greedy modularity optimization

- 1 Each vertex starts as its own community
- 2 Move vertices to neighbour's community if it increases Q
- 3 Aggregate communities into super-nodes
- 4 Repeat until Q converges

Leiden (2019): fixes a problem — Louvain can produce **disconnected communities**

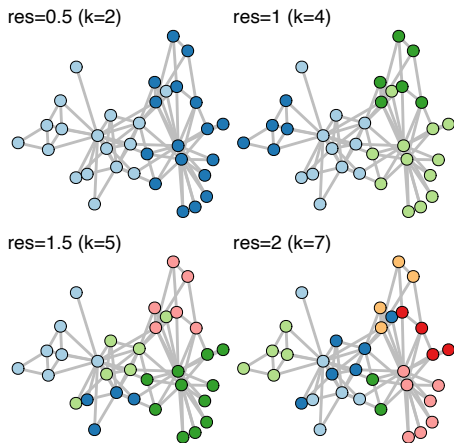
```
cl.louvain <- cluster_louvain(karate)
cl.leiden  <- cluster_leiden(karate,
                             objective_function = "modularity")
```

Louvain $Q=0.42$ Leiden $Q=0.42$



The resolution parameter matters

- Both Louvain and Leiden have a **resolution** parameter
- Higher resolution $\rightarrow$ more, smaller clusters
- Lower resolution $\rightarrow$ fewer, larger clusters
- There is no “correct” resolution — it depends on the question

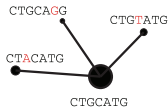


Today's lecture

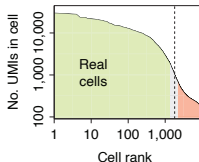
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Recall: the standard single-cell workflow

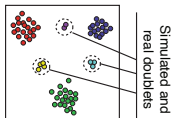
Alignment and molecular counting



Cell filtering and quality control



Doublet scoring

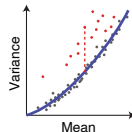


Cell size estimation

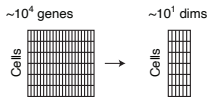
Cells	c_1	c_2	c_3
Gene ₁	2	4	20
Gene ₂	1	2	10
Gene ₃	3	6	30

Cell depth: 6 | 12 | 60

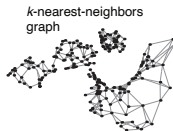
Gene variance analysis



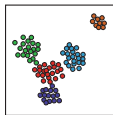
Reduction to a medium-dimensional space



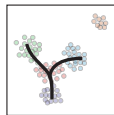
Manifold representation



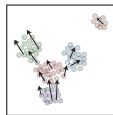
Clustering and differential expression



Trajectories



Velocity estimation



(Kharchenko2021-wr?)

Building the KNN graph

Shared neighbours graph

Clustering the PDAC data

Effect of the resolution parameter

Today's lecture

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Are your clusters real? (Grabski et al. 2023)

Even when there is **no true grouping**, standard DE after clustering produces many “significant” genes:

- The clustering algorithm forces cells into groups
- Cells assigned to the same cluster are more similar *by construction*
- Standard tests (e.g., Wilcoxon, t-test) ignore this selection step → **inflated p-values**

sc-SHC (single-cell significance of hierarchical clustering):

- 1 Hierarchically split data into two groups
- 2 Fit a **single parametric model** (negative binomial) to all cells
- 3 Simulate 100 datasets under this null model
- 4 Compare observed Wald linkage to the null distribution

The double-dipping problem (Neufeld et al. 2023)

Setup: We observe a count matrix X (cells $\times$ genes), where $X_{ij} \sim \text{Poisson}(\lambda_{ij})$

Standard pipeline (double dipping):

- Use X to estimate clusters $\hat{c}_1, \dots, \hat{c}_n$
- Use X again to test: “Is gene j differentially expressed between clusters?”
- The test does not account for the fact that clusters were chosen to make genes look different

Result: under the null (λ_{ij} does not depend on cell type), we still get many small p-values

Count splitting (Neufeld et al. 2023)

Key insight: Poisson thinning creates independent splits

If $X_{ij} \sim \text{Poisson}(\lambda_{ij})$, then for any $\epsilon \in (0, 1)$:

$$X_{ij}^{\text{train}} \mid X_{ij} \sim \text{Binomial}(X_{ij}, \epsilon), \quad X_{ij}^{\text{test}} = X_{ij} - X_{ij}^{\text{train}}$$

Then $X^{\text{train}} \perp X^{\text{test}}$ (they are **independent!**)

The recipe:

- ① Split X into X^{train} and X^{test} via Poisson thinning
- ② Cluster using X^{train} only
- ③ Test for DE using X^{test} only

This controls Type I error because the test data is independent of the clustering.

Over-clustering (DenAdel et al. 2025)

Problem: Leiden with a high resolution parameter → too many clusters → spurious DE genes

RECALL (calibrated clustering with artificial variables):

- ➊ **Augment** the expression matrix with **artificial null genes** — synthetic variables that match the data distribution but have no association with cell types
- ➋ **Cluster** cells using both real and artificial genes
- ➌ **Test for DE** between clusters, for both real and artificial genes
- ➍ Since artificial genes are known nulls, their test statistics reveal the **double-dipping effect**
- ➎ If real genes are not more significant than artificial ones → **merge** those clusters
- ➏ Repeat until clusters are well-calibrated

Takeaways

Graph-based clustering

- Modularity optimization (Louvain/Leiden) is the workhorse of single-cell clustering
- The resolution parameter controls granularity — there is no single “correct” value
- SBMs offer a principled alternative with automatic model selection

When clustering goes wrong

- Clustering then testing on the same data = **double dipping** → inflated p-values
- **sc-SHC**: build significance testing into the clustering itself
- **Count splitting**: exploit Poisson thinning to create independent train/test sets
- **RECALL**: use artificial null genes to calibrate and detect over-clustering

References

- Blondel, Vincent D, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. 2008. “Fast Unfolding of Communities in Large Networks.” *Journal of Statistical Mechanics: Theory and Experiment*.
- DenAdel, Alan, Michelle L Ramseier, Andrew W Navia, et al. 2025. “Artificial Variables Help to Avoid over-Clustering in Single-Cell RNA Sequencing.” *Am. J. Hum. Genet.* 112 (4): 940–51.
- Grabski, Isabella N, Kelly Street, and Rafael A Irizarry. 2023. “Significance Analysis for Clustering with Single-Cell RNA-Sequencing Data.” *Nat. Methods*, July.

References

- Neufeld, Anna, Lucy L Gao, Joshua Popp, Alexis Battle, and Daniela Witten. 2023. "Inference After Latent Variable Estimation for Single-Cell RNA Sequencing Data." *Biostatistics* 25 (1): 270–87.
- Newman, M E J, and M Girvan. 2004. "Finding and Evaluating Community Structure in Networks." *Phys. Rev. E Stat. Nonlin. Soft Matter Phys.* 69 (2 Pt 2): 026113.
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- Zachary, Wayne W. 1977. "An Information Flow Model for Conflict and Fission in Small Groups." *Journal of Anthropological Research* 33 (4): 452–73.